DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	CHEST TUBE INSERTION, ASSISTING AND MAINTAINING		
Department	Neonatology Department		
Applies to	All NICU Medical and Nursing Staff		
Policy Code and Version Number	DPP-NICU-015 (v2)		
Last Version Approval Date	28/1/2022	Number of Pages	5

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APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19 / 3 / 2024	7 / 4 / 2024	TRIENNIAL	18 / 3 / 2027

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1.0 PURPOSE

1.1. To re-establish negative intra pleural pressure necessary for lung re-expansion when a pneumothorax has developed.

1.1. To provide drainage of a pleural effusion or to obtain fluid for diagnosis

2.0 DEFINITION/ABBREVIATION

2.1. **Chest tube insertion**-method of inserting tube in the pleural space to facilitate evacuation of air or fluid from the pleural cavity.

3.0 RESPONSIBILITY

3.1. NICU Physician

3.2. NICU Nurses

4.0 POLICY

4.1. Insertion of the chest tube should be performed by the physician under aseptic technique, assisted by a qualified nurse.

4.2. Chest tube should be monitored for air leak and pleurovac water chamber will be monitored for fluctuation every hour.

4.3. Rubber- tipped clamps or Kelly clamp should be ready at bedside

5.0 PROCEDURE (S)

5.1. Explain procedure to parents of infants.

5.2. Secure consent from parents.

5.3. Wash hands thoroughly.

5.4. Position the child with head of bed elevated. Secure limbs with soft restrain if any.

5.5. Monitor vital signs closely. note any change in skin color. Connect to cardio-respiratory monitor and oxygen source.

5.6. Set up under water seal bottle following direction on package

5.6.1. Open saline or water container. unwrap drainage system. Fill chamber to appropriate level

5.6.2. Place funnel in tubing or port leading to suction control chamber.

5.6.3. Pour fluid into suction control port until designated amount is reached per doctors' orders, or to specific line marked on bottle, usually indicating the 20-cm water pressure level.

5.6.4. Fill water seal chamber of drainage system to the 2-cm level.

5.7. Put on sterile cap, mask, gown and gloves for both the doctor and the nurse

5.8. Open sterile equipment and place on sterile surface.

5.9. The following procedure should be done by the physician.

5.9.1. Select the site of insertion

5.9.2. Infiltrate the area with 0.5 -1% lidocaine. Make a small incision in the skin over rib just below the intercostal space where the tube is inserted

5.9.3. Using the tip of the hemostat, puncture the pleura just below the rib and spread gently.

5.9.4. Insert the chest tube through the opened hemostat following insertion of chest tube, attach to a water -seal vacuum drainage system. 5-10 cm of suction pressure is usually used.

5.9.5. Secure the tube with silk tape or sutures done by the physician.

5.9.6. Don gloves and connects drainage system to chest tube and suction source, if suction is indicated, maintaining sterility of connector ends

5.9.7. If changing drainage system, ask patient to take a deep breath, hold if, and bear down slightly while tubing is being change quickly.

5.9.8. Adjust suction flow regulator until quiet bubbling is noted in suction control chamber.

5.9.9. Call for chest x ray as ordered by the physician

5.9.10. Monitor vital signs every hour or as per physician's order, observing for signs of dyspnea restless, irritability and fever.

5.9.11. Do not elevate drainage set above level of chest.

5.9.12. Discard gloves and disposable materials.

5.9.13. Position patient for comfort, with call button within reach.

5.10. Maintain a chest tube

5.10.1. Observe water seal chamber for bubbling. Suspect an air leak if bubbling present and patient has no known pneumothorax also suspects an air leak if bubbling noted and chest tube is clamped or if bubbling is excessive.

5.10.2. Check tube connections every 1 to 2 hours (depending on amount of drainage or orders)

5.10.3. Mark drainage in collection chamber

5.10.4. Monitor drainage system for bubbling in suction control chamber.

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5.10.5. Check for fluctuation in water seal chamber with respirations

5.10.6. If the drainage slows or stops, gently "milk "the tubing in the direction of the drainage chamber as needed.

5.11. Every 2 hours monitor chest tube dressing for adequacy of tape seal and amount & type of soiling.

5.12. Assess breathe sound.

5.13. Monitor vital signs and temperature every 2 to 4 hours. Use the following trouble shooting tip in maintaining chest tube drainage.

5.14. If drainage system is turned over & water seal is disrupted, reestablish water seal & assess patient.

5.15. If drainage decreases suddenly, assess for tube obstruction (i.e., clots or kinks) & milk tubing's.

5.16. Check that gravity drainage systems & suction systems are below of patient's chest

5.17. Watch for tension pneumothorax and hemothorax.

5.18. If drainage increases suddenly or becomes bright red, take vital signs, observe respiratory status & notify doctor.

5.19. If a dressing becomes saturated, reinforce with gauze & tape securely.

5.20. If drainage system becomes broken clamp tube with Kelly clamp or hemostat & replace system immediately or place end of tube in sterile bottle of saline solutions, place bottle below level of chest & replace drainage system immediately.

Note: clamp chest tubes for no more than a few minutes (such as during system change)

5.21. Documentation:

5.21.1. System function (type & amount of drainage)

5.21.2. Time suction was initiated or system changed.

5.21.3. Patient status (respiratory rate, breath sounds, pulse oximetry, blood pressure, skin color, temperature & mental status)

5.21.4. Chest dressing status and care done

5.21.5. Drainage characteristics and care done

5.21.6. Date and time chest tube inserted

5.21.7. Name of doctor performing chest tube insertion.

6.0 EQUIPMENTS

6.1. Prepacked chest tube tray typically consists of:
- 6.1.1. Sterile towels
- 6.1.2. 4x4 gauze pad
- 6.1.3. 3-0 silk sutures
- 6.1.4. Curved hemostat's
- 6.1.5. no. 15 or 11 scalpel
- 6.1.6. Scissors

6.2. 25- gauge needle and 3 ml syringe
6.3. Needle holder
6.4. Antibiotic ointment
6.5. 1 % lidocaine
6.6. Chest tube 8F -12F Catheter
6.7. Sterile glove, mask, cap and gown
6.8. Suction - drainage system
6.9. Adhesive tape
6.10. Under-water-seal bottles
6.11. Oxygen source

7.0 ATTACHMENT/FORMS

7.1. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.2. Physician Order Form (GDOH-INP-POF-079)
7.3. Informed Consent for Invasive/Surgical or High Risk Procedure or Operation (APP-QPS-ICISHRP-021)

8.0 REFERENCES

8.1. MOH-KSA Guidelines for Neonatal Care

8.2. CBAHI Standards NICU3rd version

8.3. KFH-NICU AI-Baha

8.4. Lippincott Manual of Nursing Practice, 7th Edition by Nettina.

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

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KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

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SN

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ :مستشفى

Region: _____ :المنطقة/المحافظة

Dept./Unit: _____ :القسم/الوحدة

MRN: | | | | | | | | | | | | :رقم الملف الطبي

Name: _____ :الاسم

Nationality: _____ :الجنسية

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days :العمر

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ :تاريخ الميلاد

Gender: ☐ Male ☐ Female :الجنس

PHYSICIAN ORDER FORM

ALLERGIC TO:

DATE	TIME	PHYSICIAN ORDERS	PHYSICIAN STAMP / SIGNATURE	RN INITIAL / JOB NUMBER

MIKHWA GENERAL HOSPITAL
AL-MIKHWA, AL- BAHA, KSA

MRN: | | | | | | | | | | |

Name: _____

Age: ☐☐ Yrs/Months Sex: M ☐ F ☐

Nationality: _____

Consultant In charge: _____

Dept: _____ Unit: _____

INFORMED CONSENT FOR INVASIVE/SURGICAL HIGH RISK PROCEDURE OR OPERATION

I confirm that I have explained to the:
☐ Patient ☐ The husband/ the wife
☐ The adult son of the patient ☐ The father/legal .
☐ Paternal grandfather ☐ Elder adult sibling
The type of the procedure / operation, _____
_____
Purpose _____
Steps _____
_____
Potential complications _____
_____
Alternatives of the operation _____
_____
Risks of no treatment _____
_____
Interpreter had been summoned? ☐ Yes ☐ No

Name: _____ Signature _____
Also I documented that in the medical records and opportunity was given to him/her to ask questions related to the operation and those were answered.
Surgeon's Name: _____

Dr. who informed the patient _____

Date & Time _____

Dr. Signature: _____

بهذا اؤكد بانني شرحت للموقع ادناه:
☐ المريض ☐ الزوج / الزوجة
☐ الابن الأكبر الراشد ☐ الأب أو الوصي
☐ الجد من جهة الأب ☐ الأخ الشقيق الأكبر

نوع الإجراء/ العملية الجراحية: _____
_____
الغرض م العملية _____
الخطوات _____
المضاعفات المحتملة _____
_____
البدائل لتلك العملية الجراحية: _____
_____
والمخاطر من عدم إجرائها مثل: _____
_____
تم الإستعانة بمترجم؟ ☐ نعم ☐ لا

اسم المترجم: _____ التوقيع _____
وقد قمت بتوثيق ذلك في الملف الطبي وقد أعطيت له (ها) الفرصة لطرح أي أسئلة متعلقة بهذا الإجراء وقد قمت بالإجابة عليها.

اسم الطبيب القائم بالعملية: _____
اسم الطبيب القائم بالشرح: _____
التاريخ والوقت: _____
توقيع الطبيب: _____

By signing of this consent I acknowledge that I have been fully informed of the operation(s) steps, potential complications, alternatives of the operation and the risks of no treatment
I was given the opportunity to ask questions related to my operation(s) and all my questions have been answered and to the effect that I have agreed to undergo the operation(s).

بتوقيعي على هذا ، فإنني اقر بانه قد تم توضيح واستيعاب كافة المعلومات الخاصة بالعملية التي سوف تجرى والاحتمالات والنتائج المترتبة على تلك العملية والمخاطر من عدم إجرائها.

و قد أعطيت لي الفرصة لطرح أي أسئلة متعلقة بالعملية/العمليات، وقد تمت الإجابة بما يدعوني للموافقة على اجراء العملية المذكورة اعلاه.

Reason why the patient is unable to sign:
☐ Minor ☐ Disabled ☐ Unconscious ☐ Other
.....

سبب عدم قدرة المريض على التوقيع:
☐ قاصر ☐ معاق ☐ فاقد الوعي ☐ سبب آخر
.....

Patient/Authorized Person Data	Witness Data	بيانات الشاهد	بيانات المريض/ ماينوب عنه
Name: _____	Name: _____	الاسم _____	الاسم : _____
Signature _____	Signature _____	التوقيع : _____	التوقيع : _____
Date & Time: _____	Date & Time: _____	الوقت والتاريخ: _____	الوقت والتاريخ: _____
I.D # _____			رقم الحفيظة: _____

In-case of Emergencies

1. Dr.'s Name/Signature: _____ 2. Dr.'s Name/Signature: _____